Stochastic Processes Semester 1

Tutorial 8 Extension Questions – Solutions

1. (a) The transition matrix of the corresponding jump chain is given by

$$P^* = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{4} & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & 0 & \frac{1}{5} \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

   (b) From the transition matrix of the corresponding jumping chain we see that the classes are:

$$\begin{aligned} \{1,3,5\} &\text{ open, hence transient;} \\ \{2\} &\text{ closed, hence (since also finite) recurrent;} \\ \{4,6\} &\text{ closed, hence (since also finite) recurrent.} \end{aligned}$$

   Note that state 2 is in fact a single absorbing state.

   (c) For each state i, let $\alpha_i^{\{2\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in state } 2 | X(0) = i)$. Then, solving for these absorption probabilities via the equation $\boldsymbol{\alpha}^{\{2\}} = \boldsymbol{\alpha}^{\{2\}} P^*$, we have

$$\boldsymbol{\alpha}^{\{2\}} = \left(\frac{25}{46}, 1, \frac{29}{46}, 0, \frac{10}{23}, 0 \right).$$

   And we have

$$\alpha_i^{\{4,6\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in class } \{4,6\} | X(0) = i) = 1 - \alpha_i^{\{2\}},$$

   since there are only a finite number of states. It follows that

$$\boldsymbol{\alpha}^{\{4,6\}} = \left(\frac{21}{46}, 0, \frac{17}{46}, 1, \frac{13}{23}, 1 \right).$$

2. (a) The transition matrix of the corresponding jump chain is given by

$$P^* = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

   (b) From the transition matrix of the corresponding jumping chain we see that the classes are:

$$\begin{aligned} \{1,2,4\} &\text{ open, hence transient;} \\ \{3\} &\text{ closed, hence (since also finite) recurrent;} \\ \{5,6\} &\text{ closed, hence (since also finite) recurrent.} \end{aligned}$$

   Note that state 3 is in fact a single absorbing state.

(c) For each state i, let $\alpha_i^{\{3\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in state } 3 | X(0) = i)$. Then, solving for these absorption probabilities via the equation $\boldsymbol{\alpha}^{\{3\}} = \boldsymbol{\alpha}^{\{3\}} P^*$, we have

$$\boldsymbol{\alpha}^{\{3\}} = \left(\frac{22}{23}, \frac{21}{23}, 1, \frac{43}{46}, 0, 0 \right).$$

And we have

$$\alpha_i^{\{5,6\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in class } \{5,6\} | X(0) = i) = 1 - \alpha_i^{\{3\}},$$

since there are only a finite number of states. It follows that

$$\boldsymbol{\alpha}^{\{4,6\}} = \left(\frac{21}{23}, \frac{2}{23}, 0, \frac{3}{46}, 1, 1 \right).$$

3. The stationary distribution is $\boldsymbol{\pi} = (4/21, 23/41, 11/42)$.

4. The stationary distribution is $\boldsymbol{\pi} = (26/75, 9/25, 22/75)$.